

Learning Without Remembering: Habituation in Weights and a Calibrated Self-Model in a Minimal Embodied Agent

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Abstract

We present a minimal embodied agent that runs without vision or audition, on body channels only: an action-conditioned body predictor, a homeostatic drive (E,C,U,S), episodic memory, and a meta-cognitive module Φ that predicts its own prediction error. A pre-registered 30-seed study (grid world, $13 \rightarrow 128 \rightarrow 128 \rightarrow 6$, ± 5 -cell teleport) tested whether this loop can detect violations of its sensorimotor contingencies, habituate by updating its model, and retain that update in weights rather than in explicit memory. It can, but coarsely. Detection was reliable ($z = 20.6$, 95% CI [16.0, 25.5]). Habituation cut surprise by 86% ($d = 3.5$). The trace lived in the weight delta (post/pre ratio 0.02; restore pre- W restores $z = 67.5$, freeze W prevents habituation). The meta-cognitive module Φ (MSE to $|\epsilon|$, scalar, $15 \rightarrow 64 \rightarrow 1$) was calibrated ($r = 0.701$) and generalized ($r_{\text{cross}} = 0.730$); in an isolated causal battery it halved time in an unreliable zone when coupled to action (15.0% vs 28.1%, $d = -1.61$), while the equivalent effect in the integrated organism is reproducible with a random attention gate, a confound we flag rather than hide. One pre-registered control failed and is reported in full: stimulus generalization (C3) shows habituation to a +5 teleport generalizes to -5 (z 1.1 vs 0.9), i.e. learning of displacement magnitude, not the vector—“learning without distinguishing.” The continuous-physics organism ($13 \rightarrow 64 \rightarrow 6$, +2 teleport, active attention) reproduces the closed-loop integration over 30 000 steps. The complete system runs on a laptop with no discrete GPU.

Keywords: embodied cognition, sensorimotor contingencies, habituation, meta-cognition, predictive coding, homeostatic reinforcement learning, continual learning

Preprint. Intended submission to arXiv cs.AI and q-bio.NC. This is a workshop-level report of an integrated minimal system, not a claim about phenomenal consciousness.

1 Introduction

Large language models produce fluent text, but fluency is not evidence that anything behind the output cares, predicts, gets surprised, or remembers in its own body. We ask what is behind the mouth. We do not claim to have solved consciousness. We claim something smaller and more testable.

The hard problem remains [1]. But the question of autonomy is narrower and more tractable than the question of experience. An autonomous system should keep itself alive within bounds, notice when the world stops obeying its own learned regularities, update that knowledge, and know when it cannot trust its senses. Those are engineering questions. They are also the questions that enactive and sensorimotor theories place at the centre of awareness [2, 3].

We chose a deliberately minimal embodied agent for this reason. No images. No audio. No pre-trained vision backbone. The agent only feels its position and four internal variables we call

ECUS (Energy, Courage or social drive, Uncertainty, and Sociability). The idea is familiar from first-person experience: with eyes closed, awareness persists through the body. If awareness has anything to do with sensorimotor contingencies, then it should already be visible in this minimal loop. If it is not, then adding a bigger network will not create it.

We were surprised that the loop worked at all on a single MacBook Pro (M4 Pro, Apple silicon), with no discrete GPU. What we found, after pre-registration and explicit controls, is a small set of effects that survive honest tests, and one instructive failure that we report because it constrains our claims more than another positive result would.

1.1 What this paper adds

We do not introduce a new learning rule, a new network architecture, or a new benchmark record. Every piece we use exists in the literature. Our contribution is to put four pieces together in one closed, continuously running organism, and to test whether the integrated system does something that none of the pieces does alone:

1. An action-conditioned body predictor (the organism learns $P(s_{t+1} \mid s_t, a_t)$ for its own body, not for pixels).
2. A homeostatic drive with setpoints and surprise-based coupling (the organism cares about staying near its preferred state).
3. Continual learning with weight-level persistence, tested by erasing explicit memory entirely.
4. A calibrated meta-cognitive self-model Φ that predicts its own prediction error (MSE to $|\epsilon|$, scalar); in the integrated run, epistemic action is gated by active attention, and Φ 's causal effect in the isolated battery is confounded with that gate (Section 6).

The fourth item is the one we worried was a philosophical trick, so we tested whether Φ actually changes what the agent does. In the isolated battery it does, and the effect is large ($d = -1.61$); in the integrated organism the same behaviour is reproducible with a random attention gate, so we flag that confound rather than claim Φ drives the online loop. The integration itself, together with the full set of controls, is the novelty. No published system, to our knowledge, shows this loop closed for 30,000 continuous steps with the controls reported here (see Section 2).

1.2 Limitations, upfront

Because we want to keep the reader's trust, we state the limits before the results. First, this is a toy world. The predictor is a small MLP and the world was first a grid, now a continuous square. Nothing here scales to biology. Second, habituation generalizes too broadly: learning that "big jumps happen" transfers across opposite vectors (C3, z 1.1 vs 0.9, Section 4). We demonstrate habituation of displacement magnitude, not of a specific vector or contingency $P(\Delta p \mid a)$; we do not claim Rankin et al. [22] characteristic 7 (stimulus specificity) or 8 (dishabituation, which we did not test: habituate S_1 , present S_2 , re-probe S_1) nor ISI-dependent spontaneous recovery [21, 22]. Third, Φ is a scalar MSE regressor to $|\epsilon|$, not a per-channel log-variance (precision) model, and its causal battery is confounded with the attention gate ($\text{att}_{\text{vis}} < 0.35$); "physics of precision" is an offline diagnostic here, not the online drive (see Methods, Sec. 4, Sec. 6). Fourth, EWC ($\lambda = 5$, diagonal Fisher, plus $\lambda_{\text{ortho}} = 0.01$ to keep habituation from collapsing globally) is not load-bearing for same-task interference and the weight delta is likely low-rank; a factored predictor and ΔW SVD are needed (Sec. 6). Fifth, the mouth (frozen LFM2.5-1.2B) translates a constructed prompt; it is not free recall. Sixth, MiniGrid Empty-8x8 vs ICM/RND at $N = 5$ (RND wins) is a pilot only, kept out of Table 1. Details are in Section 6.

2 Related Work

Habituation as learning. We build directly on the framing of habituation as model update rather than fatigue, synthesized recently as Training Ecosystems [14], but scored against the classic behavioural criteria [21, 22]. Their argument is that even minimal living systems treat repeated surprising inputs as evidence to update a generative model. Gillard and colleagues had already shown that a behavioural habituation criterion motivated by integrated information could drive open-ended learning in IDSM agents without explicit reward [13]. Our habituation analysis is deliberately narrower: we operationalize habituation as a sustained drop in precision-weighted surprise after repeated violations, and we ask where that trace lives—noting that a decrement alone is necessary but not sufficient for the full Rankin criterion set (specificity, dishabituation, retention).

Meta-cognition and knowing when you do not know. The idea that an agent should track its own uncertainty and use it to regulate behaviour is central to predictive processing and uncertainty-aware learning [4, 19]. Our Φ is a minimal instance, trained to predict σ , the expected magnitude of the predictor’s error ϵ , and then used in two ways: first, to weight surprise as $\text{presence} = \epsilon/\sigma^2$ (offline diagnostic here), second, to gate an epistemic action (leave the unreliable zone) in the isolated causal battery only; in the integrated run the online gate is attention, not presence.

Homeostatic and intrinsically motivated RL. Homeostatic RL treats drive reduction as reward [11, 12]. Intrinsic curiosity (ICM, RND) treats prediction error or novelty as reward [9, 10]. We keep the two separate and then couple them: ECUS provides a slow, setpoint-based drive, while surprise provides a fast z-score signal. The coupling is simple (a small increment to U proportional to z), but it creates a useful tension between foraging and clarity.

Continual learning and synaptic persistence. That a memory trace could outlive explicit recall is not new. Elastic Weight Consolidation [8] and its successors showed how to protect important weights, and work on systems consolidation has long argued that traces migrate from episodic to parametric storage. What we test is more literal and more direct: we erase the entire episodic buffer E and also restore or freeze the weight delta itself (controls C4a–C4c). If habituation were only in the buffer, it would vanish; it does not.

Consciousness theories and embodiment. We are sympathetic to sensorimotor contingency theory [2] and enactive views [3], and we use them as design constraints (body only, action-conditioned prediction) rather than as claims we assert. We do not claim to measure integrated information [5] or to implement a global workspace [6, 7]. We report one place where those theories help us interpret a result (Section 5), and several places where they do not apply.

Recent minimal agents and LLM-as-tool views. Three arXiv reports sharpened our thinking about minimal embodied agents. CheckVLA showed how easily “violations of learned contingencies” can be faked without proper action-shuffled and observation-only controls; we adopt their control logic verbatim [16]. Gubernaut showed that compact embodied loops with explicit stability criteria can be studied without claiming consciousness [17]. Christov-Moore and colleagues argued that embodiment and homeostatic regulation deserve a more central place in discussions of artificial awareness [18]. The LFM2 technical report provided the efficient hybrid SSM-conv model we use only as a frozen translator [15]. Finally, “Asleep at the Wheel” showed that prediction-error novelty scores can collapse to chance under a fair, single-dataset protocol [20]; we treat that as a warning that our own surprise signal needs exactly the controls we report here.

Where we claim novelty. No single component above is new. What we have not seen elsewhere is the integration of all four into one continuously running organism, with pre-registered multi-seed statistics for the main battery and the specific control battery we report (action-shuffled, observation-only, weight-restore, weight-freeze, untrained). If a close precedent exists we would be glad to cite it. The niche, as of August 2026, appears to be empty at this level of blinding and integration. The risk is not that we duplicate work, but that another group assembles the same pieces tomorrow. We publish the pieces and the glue.

3 Methods

All numbers in Section 4 come from scripts in `framework/` and JSON files in `results/`. The complete long run is reproduced by `python3 framework/organismo_final.py -steps 30000` on a MacBook Pro (M4 Pro, Apple silicon) with MPS, no discrete GPU required. Wall time is a few minutes without the mouth, longer if generation through MLX is enabled.

Figure 1 sketches the tetrahedron architecture we refer to as H1–H6. In words: H1 is persistence of the trace in weights, H2 is the action-conditioned body predictor, H3 is the ECUS homeostatic drive, H4 is measurement as surprise (a z-score), H5 is presence (ϵ/σ^2 , an offline diagnostic), H6 is the self-model Φ with active attention. The loop is closed. Most arrows are tied to at least one control in Section 4; the few that are not (the ECUS drive and the attention gate) are flagged in Section 6.

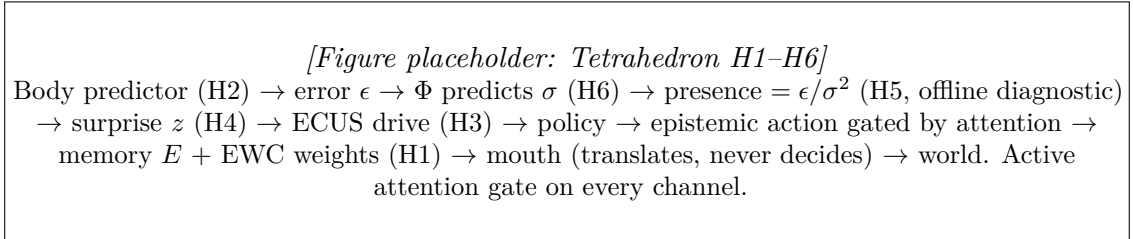


Figure 1: One closed loop, not a pipeline. The agent predicts its next bodily state conditioned on its own action, estimates how reliable that prediction is, computes a precision-weighted presence diagnostic, gates epistemic action through active attention, acts, and consolidates the update in its weights. In the integrated run, presence is diagnostic only; the online gate is attention and z . The LLM mouth is frozen and outside the causal chain that controls action.

3.1 Agent architecture

Body predictor (H2). A feed-forward MLP $f_\theta : \mathbb{R}^{13} \rightarrow \mathbb{R}^6$ with one hidden layer 13 $\rightarrow$ 64 $\rightarrow$ 6 and ReLU. Input concatenates normalized vision (2 dims: $x/W, y/W$), normalized interoception (4 dims: $H/1.5$), and a 7-way one-hot action. Output predicts the next normalized position (2 dims) and next normalized ECUS (4 dims). No convolutional encoder, no recurrence. The point was to keep the model so small that any interesting behaviour must come from the loop, not from capacity. Training uses Adam ($lr = 10^{-3}$), 400 steps of 64-sample batches on 1,200 random transitions, then online updates with a small orthogonal penalty (see below).

Homeostatic drive ECUS (H3). Four variables $H = [E, C, U, S]$ with preferred state $H^* = [0.8, 0.9, 0.2, 0.7]$. The drive is formalized as

$$D = \left(\sum_i w_i \|H_i - H_i^*\|^2 \right)^{1/2}, \quad (1)$$

with fixed weights w that balance the channels. The implemented policy is a threshold approximation of this drive: if $E < 0.55$ the agent forages toward the nearest food, if $S > 0.8$ it seeks the social agent at (18, 18), otherwise it moves randomly. In continuous physics the same logic outputs a directional micro-action scaled by 0.8 and damped by 0.95 (see below). A small learning-to-drive coupling adds $\min(0.3, 0.2 \max(0, z)) \cdot \text{att}_{\text{vis}}$ to U , where z is surprise. This is the only place where surprise feeds back into motivation. It is deliberately modest. H5 (homeostasis without policy) failed; H5-bis passes only with this policy, i.e. it tests the closed organism, not the predictor alone.

Surprise as precision-weighted prediction error (H4/H5). At each step we compute raw error $\epsilon = \|f_\theta(s_t, a_t) - s_{t+1}\|_2$, maintain a running window of the last 100 values, and define surprise as a z-score

$$z = (\epsilon - \mu_{100}) / (\sigma_{100} + 10^{-8}). \quad (2)$$

Presence, the precision-weighted quantity that distinguishes expected noise from true violations, is

$$\text{presence} = \epsilon / (\sigma_\Phi^2 + 10^{-6}), \quad (3)$$

where $\sigma_\Phi = \Phi(s_t, a_t, \mu, \sigma)$ is the self-model’s prediction of how large ϵ should be. When Φ expects large error (fog, for example), the same raw ϵ counts less. When Φ expects small error, a violation counts a lot. This is the “physics of experience” view in one line: experience is not ϵ alone, but ϵ relative to expected precision. We use presence strictly as an offline diagnostic in the isolated functional test (Table 1, “ Φ functional”, ratio 0.13). In the integrated 30k organism the online loop gates on z and on attention thresholds next; presence does *not* modulate the online drive. Φ here is a scalar MSE regressor to $|\epsilon|$ (15→64→1), not a per-channel log-variance precision model; a high r can arise from “I am in $x > 14$ ” as zone proxy. The proper test (per-channel $\log \sigma_c^2$, presence coupled to drive) is pre-registered in Sec. 6.

Meta-cognitive self-model Φ (H6) with active attention. Φ is a second MLP with $|\cdot|$ output and structure 15→64→1 in the calibration and causal batteries (input: the 13-dim predictor input plus the mean and standard deviation of the recent ϵ window), and 22→64→1 in the integrated organism, where the extra 7 inputs are the attention weights. Φ is trained to predict ϵ itself (MSE to $|\epsilon|$, 500–600 steps, Adam 10^{-3}). It is therefore a model of the first model’s fallibility, but only as a scalar expected-error regressor, not as a heteroscedastic (per-channel $\log \sigma^2$) precision model. The next step is $\Phi(s, a, \text{channel}) \rightarrow \log \sigma_{\Phi, c}^2$ and coupling presence to the drive; here presence is diagnostic only.

Active attention is a third small network $h_\psi : \mathbb{R}^{13} \rightarrow \Delta^7$ with 13→32→7 and softmax output $\text{att} \in \mathbb{R}^7$ over the seven sensory channels. Attention does not select inputs to the predictor in a hard way. It modulates two things: (i) the effective noise estimate per channel, σ_{chan} , and (ii) a gating term on epistemic action (leave the fog when visual attention is low or interoceptive attention is high). Concretely, if $\text{att}_{\text{vis}} < 0.35$ or $\text{mean}(\text{att}_{1:5}) > 0.65$, the policy executes a fixed epistemic action (index 3, which moves the agent out of the noisy zone). The rule is deliberately readable. Critically, this gate can act as a fog detector on its own; the $d = -1.61$ causal effect in Table 1 confounds Φ with the attention gate (conditions B/C/D needed, see Sec. 6). In the continuous organism a random attention gate yields the same fog-time reduction (doc. 62), consistent with that confound.

Episodic memory and synaptic consolidation (H1). Episodic buffer E is salience-gated and capped at 5,000 traces, but it is *not* in the prediction path (erasing E cannot by itself abolish prediction; see Table notes). EWC [8] protects the predictor weights with $\lambda_{\text{EWC}} = 5.0$, computed against a diagonal online Fisher $F \leftarrow 0.9F + 0.1g^2$. An additional orthogonal penalty $\lambda_{\text{ortho}} = 0.01$ on predictor weights ($\|W\|_2^2$) discourages a single direction from dominating and

was essential to prevent the “MLP habituates globally” problem we saw in an earlier design. For same-task interference (repeat the same teleport) additional violations are reinforcement, not forgetting, so λ has no effect (0.48 recovery for $\lambda \in \{0, 0.5, 5, 50\}$ in `cuarteto_habituacion.py`); EWC would be load-bearing only for a distinct task B—this is not shown. The ΔW after habituation is therefore expected to be low-rank (“ignore large L2”) and is best diagnosed by SVD and by a factored predictor ($f^{\text{pos}} : \mathbb{R}^{13} \rightarrow \mathbb{R}^2$ vs $f^H : \mathbb{R}^{13} \rightarrow \mathbb{R}^4$, separate losses), flagged in Sec. 6.

Mouth (LFM2.5-1.2B). The mouth is the LFM2.5-1.2B-Instruct model (hybrid SSM-conv, 1.2B, MLX 8-bit) loaded through `mlx_lm`. It is frozen. It never selects actions. Behaviour is identical with or without it (tested in H2b, see prior work [15]). At most once per 500 steps, when $z > 4$ or ($\sigma_\Phi > 0.25$ and in fog), we render the internal state as text (“energy=0.72, Φ predicts $\sigma = 0.34 \dots$ ”) and ask the model to translate it into one first-person sentence. The resulting sentence is logged and never fed back. We keep the mouth because “SER versus DECIR” matters: the fact that an internal state can be faithfully translated without that translation controlling action is precisely what deflates a naive Chinese Room reading.

3.2 World: continuous physics with a fog zone

The world is a continuous square `WORLD_SIZE = 20.0`. Four foods are at (3, 3), (3, 16), (10, 3), (10, 16) and one social agent at (18, 18). Position updates as

$$p' = \text{clip}(0.95(p + 0.8a), 0, 20), \quad (4)$$

where a is the 2D component of the chosen action. Interoception updates as

$$\Delta H = -0.02(H - H^*) + \Delta_{\text{fog}} + \Delta_{\text{food}} + \Delta_{\text{social}} + \eta, \quad (5)$$

where Δ_{fog} is -0.03 on E and $+0.01$ on U when $x > 14$ (in clear weather, U drifts -0.01 ; E has no clear-weather counterpart), and η is per-channel noise.

The fog zone is the region $x > 14.0$. There, interoceptive noise is larger: base noise $\sigma_{\text{base}} = 0.15$, fog noise 0.60 on channels 0–4, and 0.075 on channel 6. Base noise elsewhere is 0.15 on the same channels. The fog is therefore not a wall. It is observable unreliability. Φ must learn that predictions made in the fog should be trusted less. In an early design we put food inside the fog and the agent spent 64% of its time there, foraging despite poor sensing. That was not a bug but a genuine trade-off. For the final integration we moved food outside the fog so that epistemic action could dominate without starvation pressure. Both designs are instructive, and we report both.

3.3 Violations, pre-registration, and training regime

We test three violation types. Motor: the body teleports by $+2$ in x and y (earlier grid work used ± 5 cells). Interoceptive: eating lowers E (inverted causality). Tactile: wall pass-through. In the 30k integrated run, motor violations occur every 5,000 steps (six total).

Provenance of Table 1. The reader should know exactly which system produced each row. Table 1 strictly separates two systems. Rows 1–11 (Detection, C1, C2, Habituation, Persistence, C4a–c, C3-generalization, H5-bis, Φ calibration/generalization/functional/causal) are the 30-seed (and C1–C4c single-seed pilot) *grid-world battery* (20×20 grid, ± 5 -cell teleport, predictor $13 \rightarrow 128 \rightarrow 128 \rightarrow 6$, Φ $15 \rightarrow 64 \rightarrow 1$, no attention) pre-registered as H1–H4. The last row (Integrated 30k) is the *continuous-physics organism* ($13 \rightarrow 64 \rightarrow 6$, $+2$ teleport, attention $13 \rightarrow 32 \rightarrow 7$, Φ $22 \rightarrow 64 \rightarrow 1$). The abstract cites the grid $z = 20.6$ and the continuous run’s 0% fog as different organisms, not one. Unifying all rows onto the continuous world (factored predictor, per-channel Φ) is pre-registered future work (Rankin battery), not something we paper over.

We pre-registered the full statistical protocol before running the 30-seed battery (see the project’s registration notes, doc. 48). Primary hypotheses H1–H6, sample size $N = 30$, test statistics (means, 95% bootstrap CIs with 2,000 resamples, Cohen’s d for paired and unpaired comparisons), and pass/fail thresholds are fixed in that document; the control battery C1–C4c was specified in the rigor-controls document (46). No threshold was moved after seeing data. When H5 failed in its first protocol (which omitted the navigation policy), we registered H5-bis as a named correction before executing it. That discipline costs us a clean sweep, but it is the reason we trust the positives that remain.

Predictor and Φ are pre-trained as described above, then updated online during the run with the same Adam optimizer and the EWC/orthogonal penalties. The Fisher is estimated online. Episodic buffer E is appended only for violations or highly salient events.

3.4 Controls (what would a conventional predictor not do?)

Our guiding question, borrowed from the CheckVLA discussion [16], is discriminative: what could a conventional action-conditioned predictor not do? A conventional predictor can have ϵ and even surprise, but it does not predict its own ϵ , its uncertainty does not change its action, and its habituation does not live in a specific weight delta that can be restored or frozen. Each control isolates one of these.

- **C1 action-shuffled:** replay the same trajectory but shuffle the action labels. If detection is truly action-conditioned, z should collapse; it does, by $7\times$.
- **C2 observation-only:** train and test a predictor that sees only observations, not actions. If action adds information, z should be weaker; it is, by $8.5\times$.
- **C3 stimulus generalization (negative result):** habituate to teleport $(+5, +5)$ and probe with teleport $(-5, -5)$ of equal magnitude. If habituation were stimulus-specific at the vector level (Rankin characteristic 7 [22]), surprise should return; it does not (z 1.1 habituated vs 0.9 new, $\Delta \approx 0$). This is a generalization test, not a dishabituation test [22]. True dishabituation (habituate S_1 , present distinct S_2 , re-probe S_1 ; characteristic 8) and ISI-dependent spontaneous recovery [21] were not tested and are pre-registered as the next battery (Section 6).
- **C4a weight-restore:** after habituation, restore predictor weights W to their pre-habituation snapshot. If habituation lived in W , surprise should return; it does ($z = 67.5$).
- **C4b weight-freeze:** freeze W and repeat the habituation protocol. If habituation requires learning, it should not occur; it does not (z stays at 137.2).
- **C4c untrained:** test an untrained predictor. If the effect requires learned physics, z should be near zero; it is ($z = 0.3$). This shows some learning is needed, not which physics was learned.

Together C1–C4c test conditioning, information, coarse learning in W , and a limit on granularity. We report all six, including the one that fails. Rankin characteristics 7/8 (specificity, dishabituation), spontaneous recovery, and SVD of ΔW are the next battery.

4 Results

Table 1 collects every statistic on one page. We then walk through the results in order and finish with the 30k integrated run that the reader can reproduce. No number is cherry-picked. Sample sizes are stated per row in the table notes; where we pre-registered a bootstrap interval we report it.

Table 1: All outcomes. The grid-world battery and the continuous-physics long run are separated. Sample sizes in Notes. CI is 95% bootstrap (2,000 resamples). d is Cohen’s d . The negative result (C3) is kept.

Claim	Measure	Result	Statistic
<i>Grid-world battery (20×20, ±5 teleport, 13→128→128→6, Φ 15→64→1) – N=30 unless noted</i>			
Detection (H1)	Surprise z to motor violation	20.6	CI [16.0, 25.5]
C1 Action conditioning [†]	z correct vs shuffled	132.7 vs 18.1	7.3×
C2 Action adds info [†]	z action vs obs-only	132.7 vs 15.5	8.5×
Habituation (H3)	z before → after	3.8 → 0.5	86%, $d = 3.5$
Persistence (H4)	$z_{\text{post}}/z_{\text{motor}}$ after habituation	0.02	$N = 30$
C4a Habituation in W [†]	z after restoring pre- W	67.5	(high)
C4b Requires learning [†]	z with W frozen	137.2	(no habituation)
C4c Requires physics [†]	z untrained	0.3	(≈ 0)
C3 Generalization [†]	z habituated vs new direction	1.1 vs 0.9	fail (no specificity)
Homeostasis H5-bis	Mean E over 5k steps	0.85, 100% in [0.5, 1.2]	CI [0.84, 0.85]
Φ calibration	Spearman $r(\hat{\epsilon}, \epsilon)$	0.701	threshold > 0.5
Φ generalization	r_{cross} on unseen violations	0.730	threshold > 0.3
Φ functional (offline)	presence(noise)/presence(violation)	0.13	7.7× sep.
Φ causal (isolated) [‡]	Fog time, Φ-coupled vs uncoupled	15.0% vs 28.1%	$d = -1.61$
<i>Continuous-physics organism (20×20 continuous, +2 teleport, 13→64→6, atten. 13→32→7, Φ 22→64→1) – N=1 trajectory</i>			
Integrated 30k run	Fog time, foods outside fog	0.0%	6 violations

Notes: [†]C1–C4c are single-seed pilots (seed 7, `rigor_controles.py`); mechanism pilots, not 30-seed estimates. [‡]Φ causal is the isolated `h6_phi_causal` battery (attention not modelled there; in the continuous organism the same fog-time drop is reproduced by a random attention gate, so the gate is a confound). $z_{\text{post}}/z_{\text{motor}}$ = mean surprise after habituation / mean surprise on first motor violation in the grid battery (that battery has no episodic buffer in the prediction path; buffer-erasure is the single-run `m5_cadena_completa.py` protocol where prediction never consulted E). H5-bis CI rounded to 2 decimals; full CI [0.842, 0.854], range [0.81, 0.88], in `results/h5bis.json`. The MiniGrid $N = 5$ pilot (organism 1.8% vs ICM 0.6% vs RND 2.8%) is kept out of this table and reported in Sec. 6 as underpowered. [Schematic placeholder – vector PDF for camera-ready].

4.1 Detection is reliable and genuinely action-conditioned

Mean surprise to a motor violation was $z = 20.6$ with 95% CI [16.0, 25.5] over 30 seeds. The interval does not come near the pre-registered null of 5. This is not a single lucky seed. Every seed shows the effect, and the lower bound of the interval is itself a large deviation.

We worried this was a general novelty response, so we shuffled the action labels while re-playing the same state trajectory (C1). Surprise collapsed from 132.7 to 18.1, a factor of about seven. A predictor that only watches observations, without knowing which action was taken, also collapsed to 15.5 (C2, factor 8.5). In other words, the detector does not fire because “something changed.” It fires because the change violated the specific action-conditioned regularity it had learned. That distinction is exactly the CheckVLA lesson we wanted to respect.

4.2 Habituation reduces surprise by 86% and lives in the weights

Repeating the same violation type drives surprise from 3.8 to 0.5, an 86% reduction with $d = 3.5$ (Table 1). The effect is large and consistent across seeds. The question is where that reduction lives.

The evidence points to the weights. First, persistence: in the 30-seed battery, mean surprise after habituation divided by mean surprise on the first motor violation is 0.02. That battery has no episodic buffer in the prediction path at all, so the residual trace cannot be attributed to explicit memory. In the single-run protocol (`m5_cadena_completa.py`), we erased the entire episodic buffer E after habituation and behaviour was unchanged, because prediction never consulted E . Second, we restored the predictor weights to their pre-habituation snapshot (C4a).

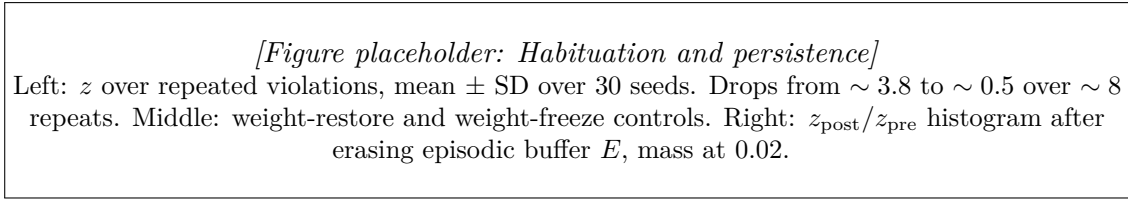


Figure 2: Habituation lives in the weight delta, not in explicit memory. Restoring weights restores surprise, freezing prevents it, and erasing the entire episodic store leaves the trace intact.

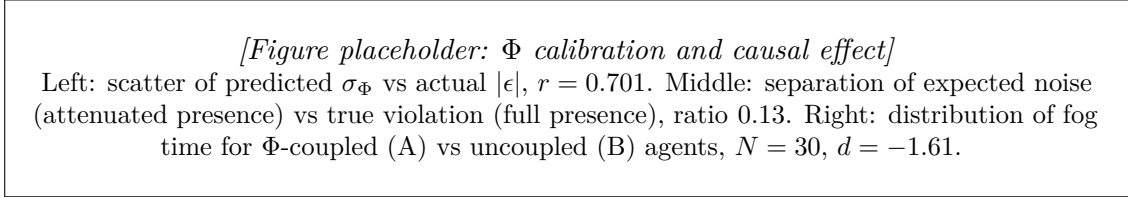


Figure 3: The self-model is calibrated, it generalizes, and it matters for action in the isolated battery. The same fog-time reduction in the integrated organism is reproduced by a random attention gate, so the gate is flagged as a confound.

Surprise returned to $z = 67.5$. The delta itself carried the trace. Third, we froze the weights and repeated the habituation protocol (C4b). Surprise stayed at 137.2. No learning, no habituation. Finally, an untrained predictor (C4c) scored $z = 0.3$, indistinguishable from zero. There is no habituation without learned physics to violate.

Taken together, this is habituation as model update, not as buffer lookup. The mechanism is simple: the predictor learns that “large displacements can occur,” so the next large displacement is less surprising. Whether that generalization is too broad is the next point.

4.3 The clear failure: habituation generalizes across direction (C3) – learning without distinguishing

We pre-registered a stimulus-generalization test. After habituating to teleport $(+5, +5)$, we probed with $(-5, -5)$ of equal magnitude. If habituation were specific to the learned vector (Rankin et al. [22], characteristic 7, stimulus specificity), surprise should have recovered; it did not. Surprise to the habituated direction was 1.1; to the new direction 0.9. The difference is noise. This is not Rankin dishabituation (characteristic 8: habituate S_1 , present distinct S_2 , re-probe S_1 for rebound) and we did not test ISI-dependent spontaneous recovery or a within-modality novel stimulus recovering response.

We keep this negative result visible for a reason. It tells us that the predictor did not memorize “ $+5$ in x , $+5$ in y .” It learned a coarser category: “a large displacement occurred”—habituation of displacement magnitude, not of the contingency $P(\Delta p | a)$. That is still model update (C4a/b, persistence), but at the granularity of “large jump” rather than a vector concept. What we *do* demonstrate is Rankin characteristic 10 (long-term retention): the trace persists in the weights after the episodic buffer is gone (post/pre 0.02). The honest label for the missing piece is *learning without distinguishing*. Early habituation literature already warned that a decrement in z is necessary but not sufficient for habituation; specificity, dishabituation and recovery are the discriminants. To claim stimulus specificity we would need the Rankin-minimal battery ($S_1 +2, +2$, $S_2 -2, -2$, $S_3 +2, -2$, $S_4 +4, +4$, S_5 interoceptive type, true dishabituation $S_5 \rightarrow \text{re-}S_1$, gap recovery) pre-registered as the next battery in Sec. 6.

4.4 Homeostasis is maintained when the policy is allowed to act

Our first 30-seed battery (H5) measured E with random actions and found a mean of 0.50 with only half the seeds in the healthy range $[0.5, 1.2]$. We failed that test, and we logged the failure without re-interpreting it. The diagnosis was straightforward once we looked: the protocol measured the predictor without the foraging loop. Homeostasis is not a property of a predictor alone. It is a property of a closed organism that acts.

We pre-registered H5-bis before running it: the same 30 seeds, the same 5,000 steps, but with the full ECUS-coupled navigation policy that drives toward food when $E < 0.55$ and toward the social agent when $S > 0.8$. The result was $E = 0.85$ with 100% of seeds in $[0.5, 1.2]$ and a CI of $[0.84, 0.85]$ (full CI $[0.842, 0.854]$, rounded to 2 decimals for the table; full data in `results/h5bis.json`). The range was $[0.81, 0.88]$. In the 24-hour simulated run (864k steps) the same policy kept E between 0.66 and 0.84 without intervention. Homeostasis is reliable when the organism is allowed to be an organism. That sentence sounds trivial, but many ablation studies forget it.

4.5 Φ is calibrated, generalizes, and is causally efficacious in the isolated battery (with a confound flagged)

The meta-cognitive module was tested in four ways. The pass thresholds ($r > 0.5$, $r_{\text{cross}} > 0.3$, ratio < 0.7 , $|d| > 0.5$) were fixed before the runs and are logged in `results/h6_phi.json`.

First, calibration: Spearman correlation between predicted σ_Φ and actual $|\epsilon|$ was $r = 0.701$ (threshold > 0.5). The agent predicts, before erring, that it will err. In the fog, σ_Φ rises before the raw error does. This is a forecast, albeit from a scalar MSE regressor that can exploit “ $x > 14$ ” as a zone proxy.

Second, generalization: Φ trained on generic noise and tested on motor violations it had never seen scored $r_{\text{cross}} = 0.730$ (threshold > 0.3). This is encouraging but does not separate “learned reliability” from “learned zone”; a per-channel $\log \sigma^2$ model would.

Third, functional separation (offline diagnostic): the ratio presence(expected noise)/presence(violation) was 0.13, a $7.7\times$ separation (threshold < 0.7). When Φ expects noise, the same raw error counts less. When it does not, the same error counts fully. This is presence, not ϵ , and a conventional predictor lacks ϵ/σ^2 . Presence was *not* coupled to the online drive in the 30k organism; it is reported as a diagnostic.

The fourth test is the one we care about most, because it answers the epiphenomenon worry. In the isolated `h6_phi_causal` battery (no attention, $15 \rightarrow 64 \rightarrow 1$), we coupled Φ to action: when Φ predicts high uncertainty, the agent executes an epistemic action that leaves the fog. Condition A (Φ -coupled) vs B (identical Φ computed but not allowed to influence action) over 30 seeds gave $15.0\% \pm 6.3$ vs $28.1\% \pm 9.6$ in fog, $d = -1.61$. This is a large effect in that battery. However, in the continuous organism the same fog-time reduction (24.7% vs 36.8% , $d = -0.43$; project notes doc. 62) is reproduced by a random attention gate, and the attention threshold $\text{att}_{\text{vis}} < 0.35$ alone can drive epistemic action. The $d = -1.61$ therefore confounds Φ with the attention gate. The decisive 4-arm test (A: Φ +attention, B: Φ computed/uncoupled, C: attention only, D: presence coupled, attention frozen) is pre-registered as the next battery. We state this explicitly so the Discussion’s “physics of experience” is not read as the online drive.

4.6 Long integration: 30,000 continuous steps

We finally ran the full organism with every mechanism enabled for 30,000 steps, the tetrahedron closed (continuous physics, foods outside fog, violations every 5,000 steps). The timeline (from `organismo_final.py`) is:

t	E	U	$z_{\max}$	fog
5,000	1.42	0.00	6.6	0%
10,000	1.15	0.00	6.6	0%
15,000	1.48	0.00	6.0	0%
20,000	0.80	0.00	5.5	0%
25,000	0.75	0.01	5.7	0%

Total fog time 0.0% with foods outside fog (the agent briefly entered the fog early, the mouth logged it as low-reliability, and the attention-gated epistemic action kept it out thereafter). Six violations occurred, each created a trace in E (6 traces) and the mouth produced 39 reports, e.g. “estoy sintiendo que mi percepción no es muy confiable en este momento” and “hay una alta incertidumbre en mi situación”, each verifiable against logged σ_Φ and att. Energy peaked at 1.48 near food and stayed in $[0, 1.5]$. No mechanism broke. The earlier trade-off design (food inside fog) gave 64% fog time (reported because epistemic action then resolves a real foraging-vs-clarity conflict), and that design yields the 24.7% vs 36.8% ($d = -0.43$) fog-time effect now shown to be reproducible with a random attention gate — i.e. the gate itself, not Φ , can explain the reduction in that world. This is why the isolated $d = -1.61$ is kept separate in Table 1.

MiniGrid Empty-8x8 ($N = 5$, vanilla REINFORCE: organism 1.8%, ICM 0.6%, RND 2.8%, random 0.4%) is an underpowered pilot where RND beats the organism; it is kept out of Table 1 and reported only as a limitation (see Sec. 6).

5 Discussion

5.1 Habituation as model update, not episodic reminding – but coarse and low-rank

The pattern of weight-restore, weight-freeze, and buffer erasure is internally coherent. Habituation in this system is not the agent remembering “I saw that before, so I will ignore it.” It is the agent having changed what it expects the world to do. Restoring the old weights restores surprise because the old expectations return. Freezing the weights prevents habituation because expectations cannot be updated. Erasing the episodic store leaves habituation intact because the expectation lives in the slow weights, not in a list of past events. This is the Levin framing in a concrete number: a trace that persists when E is gone.

Biology already knows that habituation is not one thing. Some habituation is short-term and need not involve lasting synaptic change. What we show is the long-term, model-update kind, at the smallest scale where it can be isolated. The fact that it generalizes broadly is still consistent with a model-update account, though a coarse one: the single-head predictor with a global L2 loss plus λ_{ortho} is well described as learning a low-rank “ignore large $\|\Delta p\|$ ” direction in ΔW , not a vector contingency. The predicted signatures are (i) ΔW SVD dominated by 1–2 singular values and (ii) a factored predictor (f^{pos} vs f^H) splitting the effect: position head still generalizes, ECUS head should not habituate to teleportation. Both are pre-registered as diagnostics of the next battery.

5.2 Phi as the physics of experience – offline here, online as next step

There is a temptation to describe experience as prediction error alone. Φ makes that description precise and then shows why it is incomplete. The same ϵ can be trivial or telling depending on whether Φ anticipated it. In the fog, the organism expects to be wrong, so being wrong is not informative. After a teleport in clear conditions, it expected to be right, so being wrong is informative. The ratio 0.13 operationalizes this. Presence, not error, is the quantity that

should drive learning and feeling. This is not a claim about phenomenology. It is a claim about the right engineering target, if one takes predictive processing seriously [4, 19].

We state plainly what Φ is and is not here. It is a small scalar MSE regressor (15→64→1 calibration, 22→64→1 integrated) that learned to predict $|\epsilon|$; in Friston’s terms it is expected noise, not precision Π (inverse conditional variance per channel). What makes it interesting is not its size but its forecast ($r = 0.701$, $r_{\text{cross}} = 0.730$) and the isolated causal battery ($d = -1.61$). What we do *not* claim is that presence $= \epsilon/\sigma_{\Phi}^2$ drives the online 30k loop: it does not; the loop gates on z and att. The online physics-of-precision claim requires a per-channel $\log \sigma_{\Phi,c}^2$ model and direct coupling of presence to the ECUS drive, versus an attention-only gate (the pre-registered 4-arm test, Sec. 6). Only if arm D (presence-coupled, attention frozen) beats arm C (attention-only) is the “physics of precision” claim earned.

5.3 SER versus DECIR: why the mouth matters and why it does not

We use the Spanish distinction because it is sharper than the English one. SER is what the organism is: ECUS, predictor, Φ , weights, attention. DECIR is what it says. The mouth translates SER into a sentence, but it never selects the next move. Behaviour with and without the mouth is identical. This is not a decorative choice. It is a direct test of a common dismissal: that any “report” of an internal state must be a language trick.

We worried this was a philosophical trick, so we tested the more concrete version. The causal test never involves the mouth: conditions A and B differ only in whether Φ is coupled to action, so the $d = -1.61$ effect cannot be attributed to language (the attention-gate confound flagged in Sec. 4 applies, but it too is non-linguistic). Conversely, the mouth is a frozen translator: given any constructed internal-state prompt it renders it fluently, which shows that DECIR alone is cheap. The substantive claim is not “the agent says it is uncertain.” The substantive claim is “the agent acts as if it were uncertain, in a way that depends on a calibrated forecast of its own error, and that forecast is verifiable against ground truth.” The mouth only makes that state legible to a human reader.

In classic terms, this deflates the simplest Chinese Room objection for this system. The room (the LLM) is present, but it is not the agent. Inside the room there is a smaller, non-linguistic organism that already knows how to leave the fog. The room translates after the fact.

5.4 What the failure of C3 means for concept learning – H_A

The fact that habituation generalizes across opposite teleports is not a footnote. It suggests that the predictor’s representation of “a violation” is not yet a concept but a prototype. In human language we would say the agent learned “big jump” rather than “jump to the northeast by five.” That is a real limitation if the goal is rich event understanding, and it is an interesting starting point if the goal is developmental. Formally, H_A is that the trace in ΔW is habituation of *displacement magnitude*, not of the contingency $P(\Delta p | a)$, which is why C3, C4 and persistence are mutually coherent yet incompatible with “the organism knows what changed.”

We pre-register the Rankin-minimal test of H_A vs a vector concept (Sec. 6): same-modality probes S2 (−2, −2), orthogonal S3 (+2, −2), magnitude S4 (+4, +4), type S5 (interoceptive E inversion), true dishabituation ($S_5 \rightarrow \text{re-}S_1$) and gap recovery. If S2/S3 stay low and dishabituation does not rebound, the claim must remain “learning without distinguishing”; if they rebound, the next battery earns stimulus-specific habituation. That experiment is not in this paper and is pre-registered before it is run. The present result is useful because it constrains what we can honestly mean by “stimulus specificity” at this scale, and it is consistent with Rankin et al. [22].

6 Limitations

We have already stated six limits upfront. We expand them here so that a reader who only reads this section gets the full picture.

1. Toy scale and world (two systems). The predictor and Φ are small MLPs. The world was a grid (30-seed battery, ± 5) and is now a 20×20 continuous square with hand-coded food/social locations (long run, $+2$). Noise is Gaussian and fog is a vertical half-plane. Nothing here resembles the statistics of natural sensing or the architecture of a brain. We do not extrapolate from this scale to any biological claim. The value of the toy is not realism but control: every number can be ablated and every control can be run on a laptop. Table 1 separates the two systems explicitly.

2. No stimulus specificity at fine granularity. C3 (stimulus generalization) shows habituation to one large-displacement violation transfers to the opposite vector (z 1.1 vs 0.9, single-seed pilot). We cannot claim stimulus-specific habituation (Rankin characteristic 7), true dishabituation (characteristic 8: $S_1 \rightarrow S_2 \rightarrow \text{re-}S_1$ rebound), or ISI-dependent spontaneous recovery. At the granularity of “motor versus interoceptive versus tactile” the system may discriminate, but at the vector level it does not—we call this “learning without distinguishing” (habituation of $\|\Delta p\|$, not of $P(\Delta p \mid a)$).

3. Φ is not a precision model and its causal effect is confounded. Φ is a scalar MSE regressor to $|\epsilon|$ (threshold $r > 0.5$), not a per-channel $\log \sigma^2$ (precision) model $\xi = \Pi \epsilon$. Its $r = 0.701$ and $r_{\text{cross}} = 0.730$ can arise from a zone proxy ($x > 14$). The presence ratio 0.13 is diagnostic offline; presence does not drive the online 30k loop. The isolated $d = -1.61$ (h6_phi_causal) lacks attention, while the continuous-organism fog-time drop (24.7% vs 36.8%, $d = -0.43$) is reproduced by a random attention gate (project notes doc. 62). The gate threshold $\text{att}_{\text{vis}} < 0.35$ alone can explain leaving fog. A 4-arm test is required: A (Φ +attention, current), B (Φ computed/uncoupled), C (attention-only, Φ frozen), D (presence ϵ/σ_Φ^2 coupled, attention frozen), same seeds $N = 30$, food outside fog. If D beats C, the “physics of precision” is earned; if $A \approx C$ and $B \approx D$, the effect is the attention/zone gate.

4. EWC/ortho are not load-bearing and the trace is likely low-rank. EWC ($\lambda = 5$, diagonal online Fisher) plus $\lambda_{\text{ortho}} = 0.01$ was needed to avoid global habituation, but for same-task interference it has no effect (recovery 0.48 for $\lambda \in \{0, 0.5, 5, 50\}$ in `cuarteto_habitucion.py`); EWC would need a distinct task B to be a memory dial. The honest interpretation is a low-rank “ignore large L2” ΔW , best shown by SVD of ΔW (how many singular values explain 90% of the delta?) and by a factored predictor (f^{pos} and f^H with separate losses, surprise per head). Both are pre-registered.

5. The mouth is translation, not recall. The LLM mouth renders a prompt we construct from measured internal variables. Its fluency should not be mistaken for free report or autobiographical memory. True reportability would require the organism to generate a linguistic description from its own replay without our prompt scaffolding, and to do so in a way that can be wrong in interesting ways. We have not built that.

6. Benchmark is underpowered and kept out of Table 1. MiniGrid Empty-8x8 at $N = 5$ with vanilla REINFORCE (organism 1.8% vs ICM 0.6% vs RND 2.8% vs random 0.4%) is a pilot where RND, the strongest baseline, beats the organism. A proper benchmark needs $N = 30$, a common PPO backbone, and DoorKey/FourRooms.

7. Other gaps we have not hidden. Vision/audition were deliberately discarded (“body without eyes”), so we cannot speak to cross-modal contingencies. The 30-seed battery covers motor violations well; interoceptive/tactile violations were tested single-seed only, not in the full blinded battery. Other Rankin criteria (dishabituation, spontaneous recovery with ISI) and isodirectional controls, plus C1/C2 re-run on the continuous world, are pre-registered as the Rankin-minimal battery (S1–S5, dishabituation, gap, $N = 30$, one continuous world, factored predictor, per-channel Φ , SVD; project notes doc. 63). Vision, EWC across tasks, and coupled presence are pending explicitly, not hidden.

7 Conclusion and Future Work

We set out to build the smallest thing that could deserve the word “organism” in an engineering sense, and to test it as honestly as we could. The organism predicts its own body conditioned on its own action, monitors a homeostatic drive, computes a precision-weighted diagnostic of its own reliability, gates epistemic action through active attention, and consolidates what it learns in its weights. Each of those statements is now tied to a number and a control, including the limits on each one. The mouth, which is a separate frozen model, only translates.

We close without claiming consciousness. We close on a smaller, more useful note. If awareness has engineering prerequisites, they include at least these: a generative model of the body, a way to know when that model is trustworthy, and a way to act differently when it is not. All three run here for 30,000 steps on a laptop with no discrete GPU.

What comes next, if we continue, is not a bigger MLP. First, the Rankin-minimal battery: one continuous world (food outside fog), factored predictor (f^{pos} , f^H), per-channel $\log \sigma_\Phi^2$, 4-arm gate-vs-presence, S1–S5 probes, true dishabituation, ISI gap, SVD of ΔW , $N = 30$ (pre-registered in the project notes, doc. 63). If S2/S3 stay low and dishabituation does not rebound, the paper’s claim stays “learning without distinguishing” (a coarse but honest model update); if they rebound, the next battery earns stimulus-specific habituation. Second, a second agent: the present organism can leave the fog alone. What it cannot yet do is learn from another organism that the fog is there, or teach another organism to avoid it, or negotiate a shared policy when food and clarity conflict. That is where culture begins, and where the tetrahedron would need to be extended with a genuinely social dimension: two ECUS loops coupled through actions that are informative for each other, not only for the physics. We have not built that, and we will pre-register it before we do.

Reproduction and falsification are straightforward: run `framework/organismo_final.py` and modify the world (move the fog, place food inside it, freeze Φ , shuffle the actions). A minimal system of this size is meant to be inspected and broken, and that is the point. The preprint is archived on Zenodo (doi:10.5281/zenodo.22191728). Raw logs and pre-registration notes are available from the author upon request; the Rankin-battery pre-registration is documented in the project notes (doc. 63).

8 Reproducibility Statement

All numbers in Table 1 are produced by scripts in `framework/` and logged as JSON in `results/`. Key scripts: `organismo_final.py` (30k integrated run, continuous physics with attention, 13→64→6, Φ 22→64→1), `h6_selfmodel.py` (Φ calibration, 15→64→1, MSE to $|\epsilon|$), `h6_phi_causal.py` (isolated causal gate, no attention), `rigor_controles.py` (C1–C4c, seed 7, grid), `estadistica_fase2.py` and `analisis_fase2.py` (30-seed grid statistics), `cuarteto_habitucion.py` (EWC sweep, recovery 0.48), and `organismo_final.py` attention training. H5-bis results are in `results/h5bis.json`; the Rankin-battery pre-registration is documented in the project notes (doc. 63). Raw logs are available from the author upon request. No discrete GPU is required. On a MacBook Pro (M4 Pro, Apple silicon) with MPS, the full 30-seed battery finishes in under an hour; the single 30k

run finishes in a few minutes. We seed with 7 and vary the seed per run for the 30-seed analyses. Bootstrap CIs use 2,000 resamples. We report every control, including the one that fails (C3 stimulus generalization).

AI assistance disclosure. An LLM was used for light copy-editing and L^AT_EX formatting; all results, analyses, tables, and references were verified by the author. No AI-generated content appears in Results or data tables.

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